

8.6 Early Finishers: Transformations of Quadratics and a Cubic — Solutions

1. $g(x) = -2(x + 3)^2 + 5$, parent $f(x) = x^2$.

(a) From $y = x^2$, in order:

reflection in the x -axis (the factor -2 is negative);

vertical stretch with scale factor 2;

translation 3 units left and 5 units up (vector $(-3, 5)$).

Accept the reflection and stretch described together as “vertical stretch with scale factor -2 ,” and in either order relative to each other.

(b) Vertex $(-3, 5)$; axis of symmetry $x = -3$.

(c) The parabola opens downward, so the vertex is a maximum: range $y \leq 5$ (i.e. $\{y \in \mathbb{R} : y \leq 5\}$).

(d) $g(x) = -2(x^2 + 6x + 9) + 5 = -2x^2 - 12x - 18 + 5 = -2x^2 - 12x - 13$.

(e) Downward parabola, vertex $(-3, 5)$, y -intercept $g(0) = -2(9) + 5 = -13$, so $(0, -13)$.

x -intercepts (for reference): $(x + 3)^2 = \frac{5}{2}$, so $x = -3 \pm \sqrt{\frac{5}{2}} \approx -1.42, -4.58$.

2. $y = x^2$: vertical stretch $\times \frac{1}{2} \rightarrow$ reflection in x -axis $\rightarrow$ translation $(4, -3)$.

(a) Stretch: $y = \frac{1}{2}x^2$. Reflect in x -axis: $y = -\frac{1}{2}x^2$. Translate 4 right and 3 down:

$$h(x) = -\frac{1}{2}(x - 4)^2 - 3.$$

(b) Vertex $(4, -3)$.

No x -intercepts: the parabola opens downward and its maximum value is $-3 < 0$, so the graph lies entirely below the x -axis. (Equivalently, $-\frac{1}{2}(x - 4)^2 - 3 = 0$ gives $(x - 4)^2 = -6$, which has no real solution.)

(c) No — the order does not matter. Both a vertical stretch ($\times \frac{1}{2}$) and a reflection in the x -axis ($\times (-1)$) are vertical scalings, and they multiply the output: $(-1) \cdot \frac{1}{2} = \frac{1}{2} \cdot (-1) = -\frac{1}{2}$. Multiplication is commutative, so either order yields $y = -\frac{1}{2}x^2$.

3. $g(x) = 2(x + 1)^3 - 16$, parent $f(x) = x^3$.

(a) From $y = x^3$, in order:

vertical stretch with scale factor 2;

translation 1 unit left and 16 units down (vector $\begin{pmatrix} -1 \\ -16 \end{pmatrix}$).

(b) The point of inflection is the image of the origin: $(0, 0) \mapsto (-1, -16)$.

(c) x -intercept: $2(x+1)^3 - 16 = 0$, $(x+1)^3 = 8$, $x+1 = 2$, $x = 1$, so $(1, 0)$.

y -intercept: $g(0) = 2(1)^3 - 16 = -14$, so $(0, -14)$.

(d) Increasing cubic with point of inflection $(-1, -16)$, passing through $(1, 0)$ and $(0, -14)$.

4. Challenge.

(a) Complete the square:

$$p(x) = -x^2 + 8x - 13 = -(x^2 - 8x) - 13 = -[(x - 4)^2 - 16] - 13 = -(x - 4)^2 + 3.$$

From $y = x^2$: a reflection in the x -axis, then a translation 4 units right and 3 units up (vector $\begin{pmatrix} 4 \\ 3 \end{pmatrix}$).

Vertex $(4, 3)$ (a maximum).

(b) The point of inflection of $y = a(x - h)^3 + k$ is at (h, k) , so $h = -1$ and $k = 2$.

Substitute $(1, 18)$: $18 = a(1 - (-1))^3 + 2 = a(2)^3 + 2 = 8a + 2$.

$8a = 16$, $a = 2$.

Equation: $y = 2(x + 1)^3 + 2$.